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Robustness of CLIP-fusion classifiers under naturalistic and adversarial perturbations

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Abstract

Hateful meme detectors are deployed against users who actively obfuscate, yet are usually benchmarked only on clean inputs. We audit a CLIP-fusion classifier on meme datasets under naturalistic label-preserving edits and white-box adversarial attacks using three seeds and three splits. The audit exposes two failures: text edits dominate degradation as the image branch is effectively dead (image-only AUROC 0.59 vs. a dedicated image baseline 0.63); and it is undefended against white-box PGD (AUROC < 0.01 already at $\epsilon=2/255$). A lightweight pair of training-time interventions, clean $\leftrightarrow$ perturbed KL-consistency plus per-example text-modality dropout, revives the image branch (image-only AUROC $0.59 \rightarrow 0.66$, exceeding the dedicated baseline) and raises the count of fully naturalistic-robust held-out examples by 76% at zero clean-accuracy cost. The defense is class-asymmetric: 97% of corrections are benign memes, acting as a false-positive veto.

Keywords: hateful memes, multimodal robustness, CLIP, consistency regularization, modality dropout.

1. Introduction

Automated hateful meme moderation is adversarial by construction: users could rewrite captions/ re-encode images to evade it, yet such models are reported almost exclusively on clean benchmark accuracy [1, 2]. We audit a CLIP-fusion classifier under two complementary threat models — naturalistic, label-preserving obfuscations and white-box image attacks — and ask whether a cheap training-time intervention can close the naturalistic gap without sacrificing clean accuracy. Using a controlled perturbation suite (8 text $\times$ 11 image attacks $\times$ 3 severities, plus composites, determin-

istically seeded) over three seeds and three splits, we contribute: (i) an audit showing the model is a text shortcut with a dead image branch; text edits dominate naturalistic degradation while white-box PGD breaks every recipe; (ii) a lightweight recipe (KL consistency plus text-modality dropout) that revives the image branch at zero clean-accuracy cost; and (iii) a failure analysis showing the defense chiefly prevents false positives on benign memes.

2. Related Work

The Hateful Memes Challenge implements the multimodal nature of memes with benign confounders that flip the label when a single modality changes, exhibiting a model-human gap [1]. Recent systems exploit the shared CLIP embedding space [3]: HateCLIPper cross-correlates the image and text embeddings through a feature-interaction matrix [2], and MemeCLIP adds lightweight adapters and a cosine classifier for small, imbalanced meme sets [4]. These optimise clean benchmark accuracy and do not characterise behaviour under corruption or attack. A second line asks whether such models use both modalities: cross-domain performance is often text-driven while image features are dataset-sensitive [5]. Robustness to label-preserving corruptions and white-box L_∞ attacks [6, 7, 8] is well studied for unimodal vision but rarely measured for CLIP-fusion meme detectors. We close this gap with one controlled suite that diagnoses which modality is fragile and compares clean against robustness-aware training.

3. Method

Data. We use the Hateful Memes dataset [1]: 8,500 train / 500 dev memes, plus the labelled test_seen ($n=1000$) and test_unseen ($n=2000$) splits for held-out evaluation, keeping the official split structure for comparability. Complementarily, we use the MAMI [?] and MultiOFF [9] datasets, to show that our model translates well.

Classifier. A dual-encoder CLIP-fusion model on

OpenCLIP ViT-B/32 maps the image and tokenized caption to $v, t \in \mathbb{R}^{512}$. Pixel normalization is a module inside the network, so the forward pass is differentiable with respect to raw $[0, 1]$ pixels (shared by white-box attacks and consistency training). A fusion head consumes

$$z = [t; v; |t - v|; t \odot v] \in \mathbb{R}^{2048} \quad (1)$$

where $|t - v|$ captures cross-modal disagreement and $t \odot v$ alignment, then applies $\text{LN} \rightarrow \text{Lin}(2048, 512) \rightarrow \text{GELU} \rightarrow \text{Lin}(512, 1)$. We freeze the CLIP encoders and train only the 1.05M-parameter fusion head (no gain to fine-tuning 150M params on 8.5k memes); zeroing either embedding before the head yields unimodal forward modes for ablation.

Perturbation suite. Eight text attacks (e.g. leetspeak, char deletion, typos, etc.) and eleven image attacks (e.g. Gaussian noise/blur, JPEG compression, contrast shift, etc.) each have low/medium/high presets, with medium the realistic level reachable with an off-the-shelf editor and high a stress ceiling. Cells use SHA-256-keyed per-example seeds; composite cells apply $K \in \{2, 4\}$ perturbations per sample to model a determined user.

Robustness-aware training. Each batch pairs a clean view with an on-the-fly perturbed view (perturb text/image/both with prob. 0.5/0.3/0.2). AugOnly supervises both views, $\mathcal{L}_{\text{Aug}} = \text{BCE}(y, \ell_{\text{clean}}) + \alpha \text{BCE}(y, \ell_{\text{pert}})$. KL adds consistency to the detached clean prediction, $\mathcal{L}_{\text{KL}} = \mathcal{L}_{\text{Aug}} + \beta \text{KL}(p_{\text{clean}} \| p_{\text{pert}})$ with $p = \sigma(\ell)$, $\alpha=1$, $\beta=0.5$. KLDrop additionally zeros the text embedding for a fraction p of examples, forcing a real image-only prediction. The clean baseline minimizes BCE alone, with AdamW, cosine decay, and early stopping on dev macro-F1.

White-box attacks. We probe the worst case with image-space FGSM [6] and PGD [7] (L_∞ , text fixed) over $\epsilon \in \{1, 2, 4, 8\}/255$. As the targeted remedy for this regime we implement Madry-style adversarial training and a TRADES-style [8] consistency variant; ϵ -values above 8/255 showed visible noise on the photo and were therefore ignored for the purpose of this project.

Metrics. We report AUROC (threshold-free, primary), macro-F1, and FPR/FNR (moderation has asymmetric costs). Per cell we report the robustness gap $\Delta \text{AUROC} = \text{AUROC}_{\text{clean}} - \text{AUROC}_{\text{att}}$ and attack success rate (ASR), with thresholds fixed on clean dev and reused under attack. Every recipe runs for 3 seeds (mean $\pm$ std).

Model	AUROC	macro-F1	Acc.
Majority (all-0)	0.500	0.385	0.625
Image-only (ded.)	0.650	0.601	0.615
Text-only (ded.)	0.635	0.596	0.617
Multimodal fusion	0.738	0.667	0.697
Fusion model, single-branch forward:			
image-only fwd	0.602	0.563	0.591
text-only fwd	0.612	0.570	0.588

Table 1. Clean results on held-out test_unseen.

Recipe	Clean AUROC	W-text Δ	Comp Δ	Image-only	Robust cnt/2000
clean	0.738	0.110	0.073	0.602	222
AugOnly	0.708	0.063	0.043	0.620	282
KL	0.738	0.065	0.052	0.628	352
KLDrop-p015	0.743	0.069	0.054	0.658	391
KLDrop-p050	0.712	0.049	0.035	0.666	331

Table 2. Recipes on held-out test_unseen (3-seed mean).

4. Validation

We evaluate every recipe over 3 seeds on the test sets, against a majority baseline and dedicated unimodal classifiers; image and text-only heads trained from scratch.

Audit: a text shortcut with a dead image branch. Fusion beats every unimodal baseline (Table 1), but the gain is lopsided: its image-only forward (AUROC 0.60) sits roughly 0.05 below the dedicated image baseline (0.65) — about twice the deficit of the text branch — so the head leans on text, underusing the image. Naturalistic degradation is thus text-driven. This holds on all three splits (Fig. 1a); the model is more text-fragile than the dedicated text baseline.

Defense: reviving the image branch at zero clean cost. The dropout sweep (Fig. 1b) shows image-only AUROC rises monotonically with p , peaking at 0.668 ($p=0.25$), while multimodal AUROC stays within seed noise through $p \leq 0.25$. $p=0.15$ (KLDrop-p015) is the default, and $p=0.50$ is a robustness-heavy alternative. KLDrop redistributes capacity rather than adding signal (Fig. 1a): it lifts its image-only forward to or above the dedicated baseline on every split, leaving text-only unchanged. Table 2 gives the held-out trade-off: KL preserves clean AUROC, KLDrop-p015 even nudges it up while improving every robustness column, and the count surviving all 57 naturalistic cells rises $222 \rightarrow 391$ (+76%). Composites are the realistic threat: two stacked text edits cost the clean model $\Delta \text{AUROC} = 0.073$ (medium) to 0.111 (high) on composite_2text — on par with its worst single cell, yet reachable with everyday edits — and KLDrop roughly halves the medium gap ($0.073 \rightarrow 0.035$, p050).

Why it works: a class-asymmetric false-positive veto.

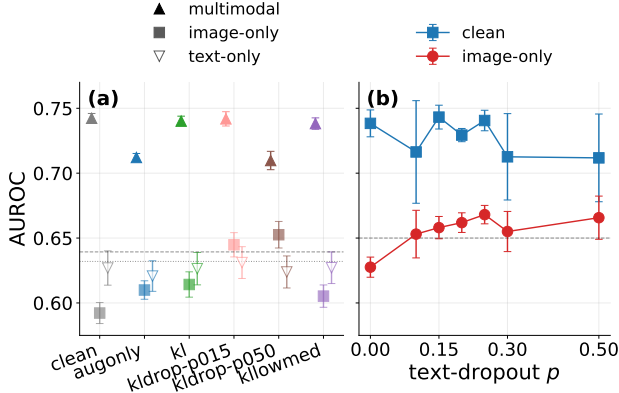


Figure 1. (a) Per-branch forward AUROC (mean over 3 splits $\times$ 3 seeds). (b) Dropout sweep (test_unseen).

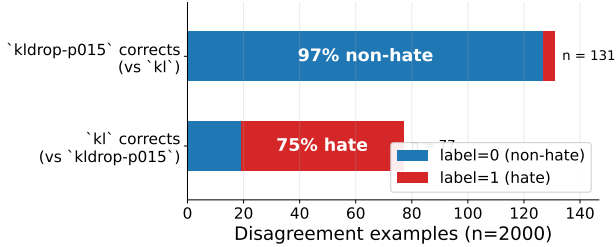


Figure 2. Per-example disagreements between KLDrop-p015 and KL on test_unseen ($n=2000$), split by gold label.

A per-example analysis on test_unseen reveals the mechanism (Fig. 2): where KLDrop-p015 and KL disagree, 97% of KLDrop’s corrections are on benign memes (95% CI [94, 99], $n=131$) vs. 75% on hateful memes for KL. The revived image branch vetoes the text branch when an edited caption looks superficially hateful but the image does not support it — the over-moderation failure that erodes trust. The choice is thus deployment-conditional: false-positive-sensitive contexts prefer KLDrop, recall-sensitive ones prefer KL; the direction holds on all three splits.

Adversarial Results. Running adversarial PGD attacks on the CLIP model trained only on clean images collapses AUROC from 0.73 to < 0.01 (ASR > 0.98) at $\epsilon=2/255$ and to 0 at $\epsilon \geq 4/255$ (see Table 3). Adversarial training restores PGD AUROC (0 $\rightarrow$ 0.61 at $\epsilon=8/255$) but, since it requires unfreezing the vision encoder, trades clean AUROC (0.74 $\rightarrow$ 0.62). Considering this white-box is unrealistic, the score is overwhelmingly positive. Other adversarial attacks such as gray-box or black-box will be much closer in performance to clean rather than white-box.

Model	Clean AUROC	Natural AUROC	PGD AUROC
Clean	0.738	0.706	< 0.01
Naturalistic	0.713	0.699	< 0.01
Adversarial	0.616	0.602	0.610
Combined	0.614	0.602	0.610

Table 3. Clean, naturalistic (mean over all cells) and white-box PGD (L_∞ , $\epsilon=8/255$) AUROC on held-out test_unseen (3-seed mean, unified trainer). Unfreezing the vision encoder (Adversarial, Combined) buys PGD robustness (0 $\rightarrow$ 0.61) at a clean/naturalistic cost; the combined model, trained against both threats, matches the adversarial model’s white-box robustness.

Combined model. Naturalistic training alone still collapses under white-box PGD (Table 3), so we unite both defenses in a single model trained jointly on three views per batch — clean, naturalistic-perturbed, and PGD-adversarial — under the composite loss $\mathcal{L}_{\text{comb}} = \mathcal{L}_{\text{KL}} + \delta \text{BCE}(y, \ell_{\text{adv}})$ ($\delta=1$, KLDrop text-dropout in the forward, ℓ_{adv} the logit of the PGD view), unfreezing the vision encoder to absorb pixel-space attacks. On held-out test_unseen (3 seeds) it inherits both: PGD AUROC at $\epsilon=8/255$ holds at 0.61 (vs. 0.00 for the clean and naturalistic-only models) at the lowest naturalistic ASR (0.09), trading clean AUROC (0.61 vs. 0.71) — the cost of the unfrozen encoder. A single model can thus resist user obfuscation and white-box attack at once.

Applicability to other Datasets. To validate our results further, we applied our training to MAMI, which showed very similar test results to the Facebook dataset. We also used our model to predict MultiOFF and results were also great.

Limitations..

5. Conclusion

A CLIP-fusion hateful meme detector is a text shortcut with a dead image branch — fragile to the text edits a real user applies, and undefended against white-box PGD. Our recipe (KL consistency plus text-modality dropout) revives the image branch above a dedicated baseline and raises held-out naturalistic robustness by 76% at zero clean-accuracy cost, acting as a class-asymmetric false-positive veto for benign content; the white-box regime remains open for adversarial training. Hateful meme detectors should thus be reported with a modality-resolved robustness audit, not clean accuracy alone, since the dominant failure — over-flagging benign speech after a trivial edit — is invisible on clean benchmarks. These are research prototypes, not moderators; we release the suite for defensive evaluation.

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